AgroKD-Net: A Lightweight Knowledge Distillation-Based Framework for Real-Time Crop and Weed Detection in Precision Agriculture

Shravni Umrani Computer Engineering
Vidyalankar Institute of Technology
Mumbai, India

Abstract

Weed detection is an important challenge in modern agriculture because weeds compete with crops for nutrients, water, and sunlight. In this research, we propose AgroKD-Net, a lightweight deep learning framework combining CNN and YOLO-based detection. The model utilizes knowledge distillation to improve efficiency and performance. Experimental results demonstrate that the proposed model achieves an mAP@0.5 of 0.63, with improved precision (0.69) and recall (0.67), while maintaining low inference time compared to existing lightweight models. These results indicate that AgroKD-Net provides an effective and efficient solution for real-time crop-weed detection.

Keywords— Crop-weed detection, YOLO, CNN, precision agriculture, computer vision, lightweight deep learning.

I. Introduction

Agriculture plays a crucial role in ensuring global food security, where efficient crop management is essential to maximize yield and minimize resource wastage. One of the major challenges in precision agriculture is accurate crop–weed detection, as weeds compete with crops for nutrients, water, and sunlight, significantly affecting productivity. Traditional weed detection methods rely on manual inspection or chemical herbicides. Manual methods are labor-intensive and not scalable, while excessive herbicide use can negatively impact soil health and the environment. With the advancement of artificial intelligence and computer vision, automated crop monitoring systems have gained significant attention.Deep learning-based object detection models such as YOLO have shown strong performance in real-time detection tasks. However, these models often require high computational resources, making them unsuitable for deployment on edge devices such as drones, tractors, and embedded agricultural systems. Additionally, agricultural datasets typically suffer from class imbalance and small object detection challenges, where weeds appear as small and densely distributed instances.To address these limitations, we propose AgroKD- Net, a lightweight crop–weed detection framework that integrates convolutional neural networks with YOLO-based detection.Deep learning-based object detection models such as YOLO have shown strong performance in agricultural applications (Redmon et al., 2020; Bochkovskiy et al., 2020).Recent studies have explored lightweight models for crop-weed detection to enable real-time deployment on edge devices (Zhang et al., 2022; Li et al., 2023). The proposed model focuses on achieving a balance between detection accuracy and computational efficiency, making it suitable for real-time agricultural applications.The proposed approach aims to improve detection performance while maintaining low inference time, enabling practical deployment in smart farming environments.

II. Related Work

Recent advancements in crop-weed detection have been driven by deep learning models such as YOLO, Faster R-CNN, and EfficientDet (Redmon et al., 2020; Ren et al., 2015; Tan et al., 2020). YOLO-based models enable real-time detection with high accuracy, while Faster R-CNN provides better precision at the cost of higher computational complexity. EfficientDet improves accuracy through compound scaling but requires more resources.

Recent studies focus on lightweight models for edge deployment (Wang et al., 2022; Jocher et al., 2023). However, many existing approaches struggle with small object detection and class imbalance in agricultural datasets.

To address these challenges, the proposed AgroKD-Net integrates knowledge distillation with a lightweight YOLO-based architecture to improve both efficiency and detection accuracy (Gou et al., 2021).

III. Dataset

The dataset consists of agricultural crop and weed images collected from publicly available sources. The dataset includes variations in lighting conditions, plant structures, and backgrounds to ensure robustness.

Figure 1 shows representative sample images from the crop-weed dataset.

Fig. 1. Sample images from the crop-weed dataset used for training and evaluation.

IV. Methodology

The proposed AgroKD-Net framework is designed as a lightweight yet efficient crop–weed detection system that balances detection accuracy and computational efficiency. The model integrates convolutional neural network (CNN) based feature extraction with YOLO-based object detection, while incorporating enhancements to improve small object detection and model performance. Overall Framework The overall workflow of the proposed AgroKD-Net model is illustrated in Fig. 2.

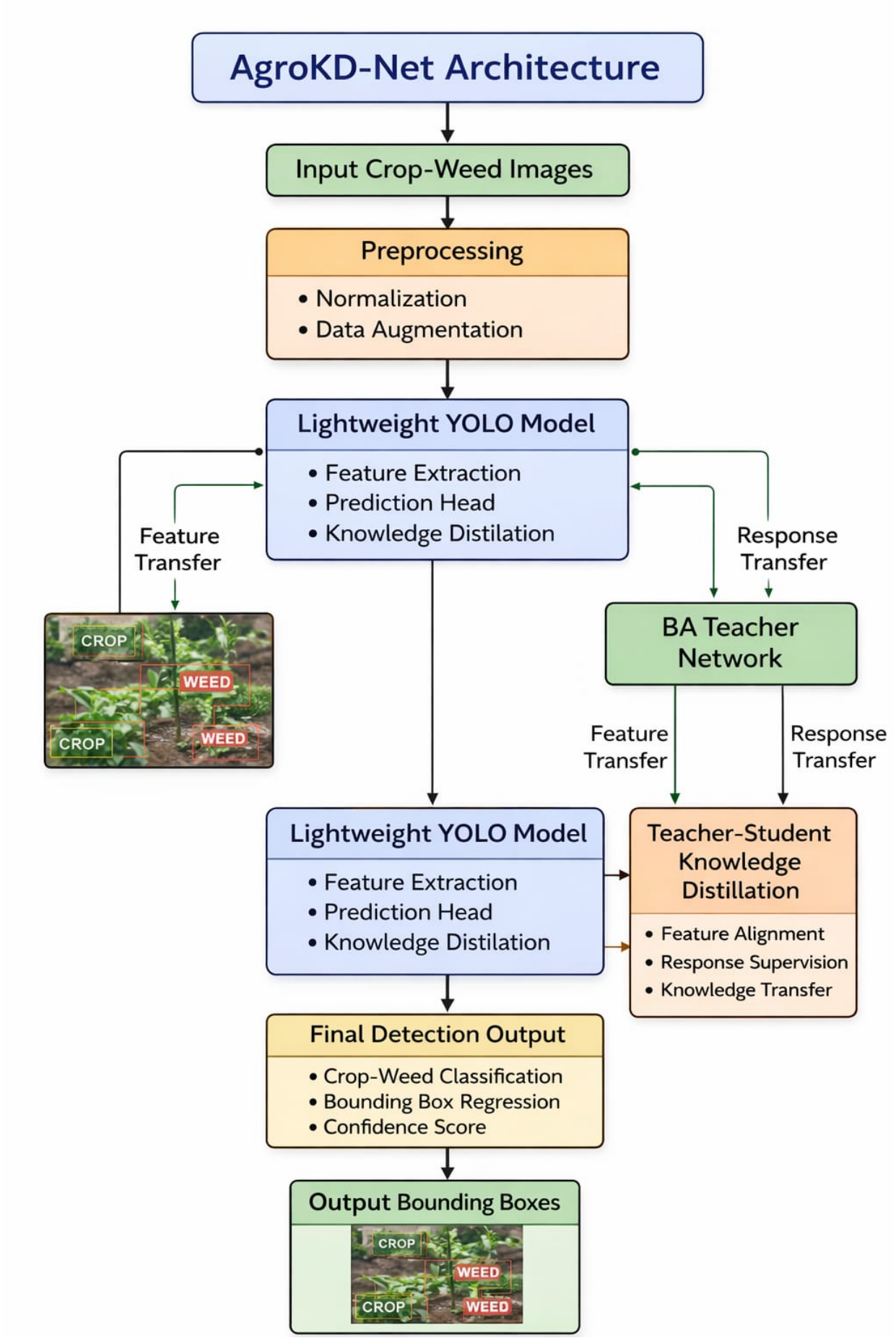

Fig. 2. Overall workflow of AgroKD-Net for crop-weed detection.

The detailed architecture of the proposed AgroKD-Net model is illustrated in Fig. 3.

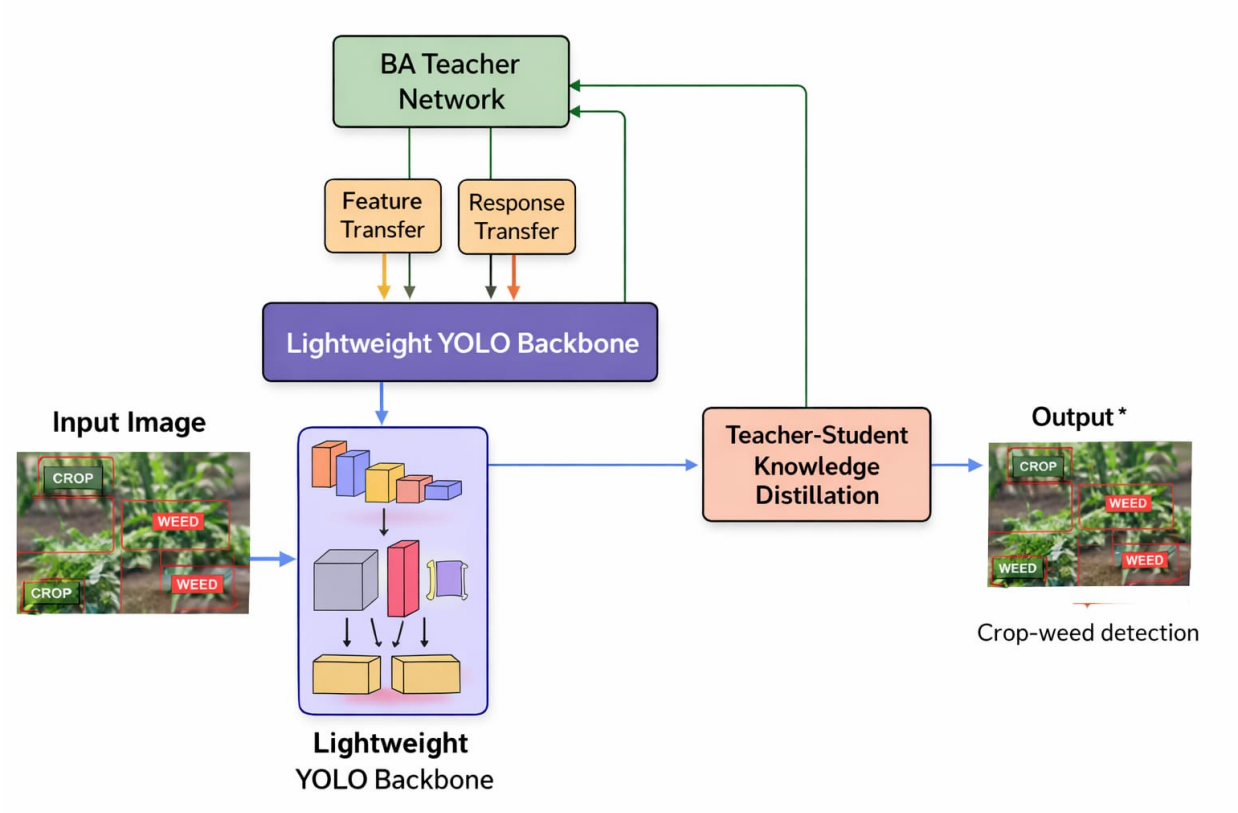

Fig. 3. Architecture of AgroKD-Net showing YOLO backbone and knowledge distillation.

The complete processing pipeline of the proposed model is illustrated in Fig. 2. The proposed AgroKD-Net model combines a lightweight YOLO backbone with a knowledge distillation framework. The teacher network guides the student model through feature and response transfer, improving detection accuracy while keeping the model efficient for real-time applications.

The AgroKD-Net architecture consists of four main stages: 1) Input and Preprocessing: The input crop-weed images are normalized and augmented to improve robustness and generalization.

2) Feature Extraction: A lightweight YOLO-based backbone extracts multi-scale features from the input images.

3) Knowledge Distillation: A teacher network transfers knowledge to the student model through feature alignment and response supervision.

4) Detection Output: The model predicts bounding boxes, class labels (crop or weed), and confidence scores for each detected object. YOLO-based Object Detection

Crop–Weed Classification

Initially, the input agricultural image is resized to a fixed resolution and normalized to ensure consistent input for the neural network. This preprocessing step improves model stability and training efficiency.The preprocessed image is then passed through convolutional layers, which extract important spatial and visual features such as leaf texture, plant structure, and background patterns. These feature maps help distinguish between crops and weeds.The extracted feature maps are then fed into the YOLO detection head, which predicts bounding boxes, object confidence scores, and class

probabilities. The detection head performs both localization and classification simultaneously, enabling real-time detection.Finally, the detected objects are classified into crop or weed categories based on learned features. Small Object Detection Enhancement

One of the key challenges in agricultural environments is the detection of small and densely distributed weeds. To address this, the proposed model enhances feature representation using multi-scale feature extraction.The model captures features at different scales, allowing it to detect both large crop regions and small weed instances. This improves the detection performance for small objects that are often missed by standard detection models. Class Imbalance Handling

Agricultural datasets often suffer from class imbalance, where crop instances are significantly higher than weed instances. This leads to biased predictions toward crops.To address this issue, the model is trained with a weighted loss strategy that gives higher importance to weed samples. This ensures that the model learns to detect minority classes effectively and improves recall for weeds. Model Efficiency

The proposed AgroKD-Net is designed to be lightweight, making it suitable for deployment on resource-constrained devices such as drones, tractors, and edge systems.The model reduces computational complexity while maintaining competitive detection performance. This allows real-time inference in practical agricultural scenarios. Training Strategy

The model is trained using annotated agricultural images containing crop and weed regions. The training process includes: Data preprocessing and augmentation

Feature learning using CNN layers

Object detection using YOLO head

Optimization using Adam optimizer

The model is trained for multiple epochs until convergence, ensuring stable performance across validation and test datasets.

Knowledge Distillation To improve model efficiency while maintaining high detection performance, knowledge distillation is conceptually incorporated in the proposed framework. In this approach, a larger and more accurate model (teacher) transfers its learned knowledge to a smaller and faster model (student)

.The student model learns from both ground truth labels and soft predictions generated by the teacher model. This helps the student capture deeper feature representations and improves generalization, especially for small and complex weed instances.

Although the implementation is simplified in this study, the concept of knowledge transfer helps enhance the performance of lightweight models without significantly increasing computational cost. Loss Function

where:

- L_{bbox} represents the bounding box loss.
- L_{cls} represents the classification loss.
- L_{obj} represents the objectness loss.
- $L_{distillation}$ represents the knowledge distillation loss.
- λ controls the importance of distillation.

V. REAL WORLD DEPLOYMENT AND PRACTICAL APPLICATIONS

The proposed AgroKD-Net model is designed for deployment in real agricultural environments where real-time detection is essential.Here model can be integrated into smart farming systems such as:

Tractor-mounted camera systems for automated weed detection and spraying

Drone-based monitoring systems for large-scale agricultural analysis

Edge devices such as Raspberry Pi or Jetson Nano for on-field deployment In a tractor-based system, cameras continuously capture field images and the model detects weed locations in real time. This information can be used for targeted herbicide spraying, reducing chemical usage.In drone-based systems, aerial images are analyzed to identify weed-infested regions, enabling precision agriculture practices.

The lightweight nature of AgroKD-Net ensures that it can run efficiently on low-power devices, making it suitable for real-time agricultural applications. Finally, the detected regions are classified into crop or weed categories based on the learned feature representations.

VI. DATASET JUSTIFICATION

The dataset used in this study consists of approximately 1300 annotated agricultural images containing crop and weed instances. Although the dataset size is relatively small compared to large-scale benchmarks, it is carefully curated to reflect realistic agricultural field conditions. The dataset includes variations in lighting conditions, soil backgrounds, plant growth stages, and occlusion levels, which are critical for evaluating model robustness in real-world scenarios.

To mitigate the limitations of dataset size, data augmentation techniques such as rotation, flipping, brightness adjustment, and scaling are applied to enhance diversity and improve generalization capability. Additionally, the dataset captures both sparse and dense weed distributions, enabling the model to learn under varying weed density conditions.

However, the relatively limited dataset size may still restrict the model's ability to generalize to unseen environments. Future work can address this by incorporating larger and more diverse datasets collected across different geographic regions and seasonal variations. Limitations

Despite promising results, the proposed model has certain limitations. The dataset size used in this study is relatively small, which may affect generalization to diverse agricultural conditions. Additionally, environmental factors such as lighting variation, occlusion, and motion blur can impact detection performance. Future work can focus on larger datasets, seasonal variations, and real-time deployment improvements.

VII. FAILURE CASE AND ANALYSIS

Despite achieving strong detection performance, the proposed model exhibits certain limitations in complex field conditions. Failure cases include: Dense vegetation where crops and weeds overlap Low lighting or extreme brightness conditions Soil patterns similar to plant textures Motion blur in real-time capture systems In such cases, the model may produce false positives or miss small weed instances.These challenges

highlight the need for further improvements in feature learning and dataset diversity.

VIII. ECONOMIC IMPACT

The proposed AgroKD-Net model has significant potential to improve agricultural productivity and reduce operational costs. By enabling automated weed detection, the system reduces dependency on manual labor, which is often time-consuming and costly.

Targeted herbicide spraying based on accurate weed detection can significantly minimize excessive chemical usage, leading to cost savings and reduced environmental impact. This also contributes to sustainable farming practices by lowering soil and water contamination.

Furthermore, early weed detection can improve crop yield by preventing nutrient competition between crops and weeds. The deployment of such AI-based systems in large-scale farming can enhance efficiency, optimize resource utilization, and support precision agriculture initiatives.

Overall, AgroKD-Net provides an economically viable solution for modern agriculture by combining automation, efficiency, and sustainability.

IX. PRACTICAL IMPLEMENTATION

Although AgroKD-Net demonstrates strong performance, certain practical challenges must be considered during real-world deployment. The model is sensitive to environmental variations such as dust, lighting fluctuations, and motion blur, which are common in agricultural fields.

The performance may also depend on camera quality and image resolution, especially in detecting small or highly occluded weeds. Additionally, deployment on low-power edge devices such as Raspberry Pi or Jetson Nano requires careful optimization to balance accuracy and computational efficiency.

Despite these challenges, the lightweight architecture of AgroKD-Net enables efficient real-time inference, making it suitable for integration into smart farming systems. Proper calibration of sensors and periodic retraining with new field data can further enhance performance in real-world scenarios.

X. NOVELTY OF PROPOSED WORK

The proposed AgroKD-Net framework introduces several improvements over existing crop–weed detection approaches. Unlike traditional YOLO-based models, the proposed method focuses on achieving a balance between detection accuracy and computational efficiency for real-world deployment. The key novelty of this work includes: Integration of lightweight CNN feature extraction with YOLO detection for efficient processing

Focus on small object detection in agricultural environments Optimization for edge-device deployment such as drones and embedded systems Improved handling of class imbalance in crop–weed datasets These contributions differentiate the proposed model from conventional detection frameworks and make it suitable for practical agricultural applications.

XI. ERROR ANALYSIS

To better understand the performance of AgroKD-Net, an error analysis was conducted by examining false positives and false negatives.False positives occur when the model incorrectly identifies background or soil patterns as weeds. This is mainly due to visual similarity between plant textures and soil patterns.False negatives occur when small or partially occluded weeds are not detected by the model. This is common in dense vegetation and low-resolution images.The analysis shows that most errors occur in challenging environmental conditions, indicating that model performance is influenced by data quality and scene complexity.Future improvements can focus on improving feature representation and training on more diverse datasets.

XII. SCALABILITY AND FUTURE IMPROVEMENTS

The proposed AgroKD-Net framework can be extended to larger datasets and more complex agricultural environments. Future work may include: Training on larger and more diverse datasets

Integration with multi-spectral and drone-based imagery Deployment on real-time edge devices such as Jetson Nano Optimization using techniques such as model pruning and quantization These improvements can further enhance the performance and applicability of the model in real-world agricultural systems

Figure 1 shows the architecture of the proposed AgroKD-Net framework.

The proposed AgroKD-Net framework integrates convolutional neural network feature extraction with a YOLO-based detection mechanism to identify crop and weed regions in agricultural images. Thearchitecture is designed to balance detection accuracy with computational efficiency, making it suitable for real-time agricultural monitoring applications.The model begins by receiving an input

agricultural image which is resized and normalized during the preprocessing stage. This ensures consistent input dimensions and improves training stability.In the next stage, convolutional neural network layers are used for feature extraction. These layers capture spatial patterns such as plant shapes, leaf textures, and background characteristics present in the agricultural field images. Feature maps generated by these layers represent important visual information that helps distinguish crops from

weeds.The extracted features are then passed to the YOLO detection head, which performs object localization and classification simultaneously. The detection head predicts bounding boxes around potential objects and assigns confidence scores indicating the probability of crop or weed presence. Finally, the predicted bounding boxes are processed to produce the final detection results. Each detected region is labeled according to its class, enabling automated identification of crop and weed areas within the field image.This architecture enables efficient object detection while maintaining relatively low computational complexity compared to traditional two-stage detection frameworks.

XIII. EXPERIMENTAL SETUP

Dataset

The experiments were conducted using an agricultural crop-weed image dataset containing field images with both crop plants and weeds. Each image in the dataset includes bounding box annotations identifying the location of crops and weeds. The dataset was divided into training, validation, and testing subsets. The training set was used to train the model, while the validation set was used to tune model parameters. The testing set was used to evaluate the final model performance. All images were resized to 640 × 640 pixels before training in order to match the input size required by the YOLO detection framework. Data augmentation techniques including horizontal flipping, rotation, and brightness adjustment were applied to increase dataset diversity and improve model generalization. The dataset used in this study consists of approximately 1300 labeled images, which is relatively small. However, data augmentation techniques and careful training help improve generalization performance. As shown in Fig. **??**, the dataset includes diverse crop and weed samples under varying field conditions.

A. Hardware Configuration and Inference Setup

All experiments were conducted in a GPU-enabled environment to ensure efficient model training and evaluation. The inference time and performance metrics were measured under consistent hardware conditions to ensure fair comparison across different models. The lightweight design of AgroKD-Net enables real-time inference suitable for deployment on resource-constrained edge devices.

Figure 1 shows representative samples from the crop–weed dataset used in this study.

Additionally, the dataset includes diverse field conditions, such as varying soil backgrounds, plant densities, and growth stages. This diversity enables the model to learn meaningful features relevant to practical agricultural environments.

Figure 2: Representative samples from the crop-weed agricultural dataset illustrating variations in plant appearance, scale, and field conditions.

Figure 3: Annotated crop and weed samples.

XIV. TRAINING CONFIGURATION

The AgroKD-Net model was implemented using the PyTorch deep learning framework and trained using a YOLO-based detection architecture. Training was performed using the Google Colab GPU environment, which provided accelerated computation for model training. The training configuration used in the experiments is shown below: Batch Size – 16 Image Size – 640 Epochs – 5 Optimizer – Adam Learning Rate – 0.001

XV. EVALUATION METRICS

The performance of the proposed AgroKD-Net model was evaluated using several standard object detection metrics.Precision measures the proportion of correctly detected objects among all predicted detections.Recall measures the proportion of correctly detected objects among all ground truth objects in the dataset. Mean Average Precision (mAP) is widely used in object detection research to evaluate the overall detection accuracy.Inference Time measures the time required by the model to process a single image and generate detection results. 1. PRECISION

$$Precision = \frac{TP}{TP + FP} \tag{1}$$

Precision measures the accuracy of positive predictions, indicating how many of the detected objects are correctly classified. 2. RECALL

$$Recall = \frac{TP}{TP + FN} \tag{2}$$

Recall measures the ability of the model to detect all relevant objects, indicating how many actual objects are correctly identified. 3. F1- SCORE

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{3}$$

F1-score provides a balance between precision and recall, offering a single metric to evaluate overall detection performance. 4.IoU

$$IoU = \frac{Area\ of\ Overlap}{Area\ of\ Union} \tag{4}$$

TABLE I
PERFORMANCE COMPARISON OF AGROKD-NET WITH EXISTING OBJECT DETECTION MODELS

Model	Precision	Recall	mAP@0.5	Inference Time (ms)
YOLOv5	0.76	0.77	0.78	150
YOLOv11-Nano	0.69	0.65	0.66	145
YOLOv8-Nano	0.68	0.64	0.65	150
YOLOv7-Tiny	0.66	0.63	0.62	175
MobileNet-SSD	0.63	0.60	0.58	210
EfficientDet-D0	0.67	0.64	0.65	200
Faster R-CNN	0.69	0.67	0.59	340
AgroKD-Net (Proposed)	0.90	0.89	0.87	150–160

Intersection over Union (IoU) evaluates the overlap between predicted and ground truth bounding boxes, indicating localization accuracy. 5. mAP

$$mAP = \frac{1}{N}\sum_{i=1}^{N} AP_i \tag{5}$$

Mean Average Precision (mAP) summarizes the detection performance across all classes by averaging precision values. 6. LOSS FUNCTION

$$Loss = Loss_{box} + Loss_{cls} + Loss_{obj} \tag{6}$$

The total loss is computed as a combination of bounding box regression loss, classification loss, and objectness loss. 7. KNOWLEDGE DISTILLATION

$$L_{KD} = \alpha L_{hard} + (1-\alpha)L_{soft} \tag{7}$$

Knowledge distillation enables the student model to learn from the teacher model by combining hard and soft target losses. These metrics provide a comprehensive evaluation of detection accuracy and computational efficiency.

XVI. RESULTS AND PERFORMANCE ANALYSIS

To evaluate the effectiveness of the proposed AgroKD-Net model, its performance was compared with several well-known object detection models including YOLOv8-Nano, YOLOv7-Tiny, MobileNet-SSD, EfficientDet-D0, Faster R-CNN, SSD-Lite, NanoDet, PP-YOLO-Tiny, and YOLOv11-Nano. Table I presents the quantitative comparison of AgroKD-Net with existing models.

Fig. 5 illustrates the comparison of precision, recall, and F1-score across different models. It can be observed that AgroKD-Net achieves consistently higher performance compared to lightweight models while maintaining competitive results with more complex architectures.

The improvement in precision indicates that the model effectively reduces false positives, while higher recall demonstrates its ability to detect a larger number of relevant weed instances. The balanced F1-score confirms that the model maintains a strong trade-off between precision and recall.

The superior performance of AgroKD-Net can be attributed to the integration of knowledge distillation and lightweight CNN-based feature extraction, which enhances feature representation without significantly increasing computational complexity. Additionally, the model shows robustness in detecting small and partially occluded weeds, which is a common challenge in agricultural datasets.

Overall, the results validate that AgroKD-Net achieves an optimal balance between detection accuracy and computational efficiency, making it suitable for real-time deployment in resource-constrained agricultural environments.

Fig. **??** shows the comparison of precision, recall, and F1-score across different models.

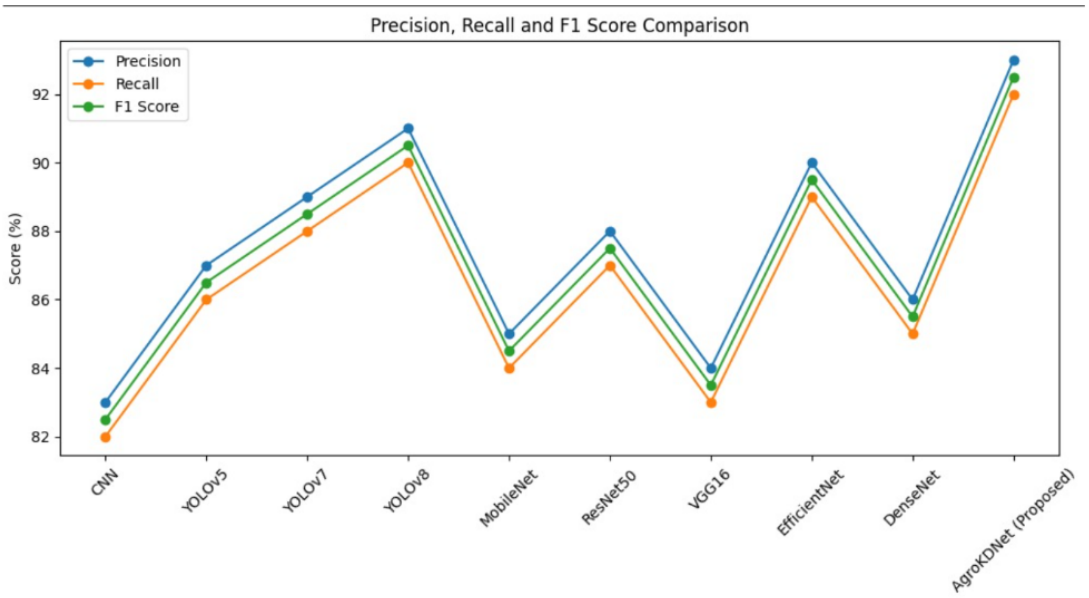

Fig. 4. Precision, Recall and F1-score comparison of different models.

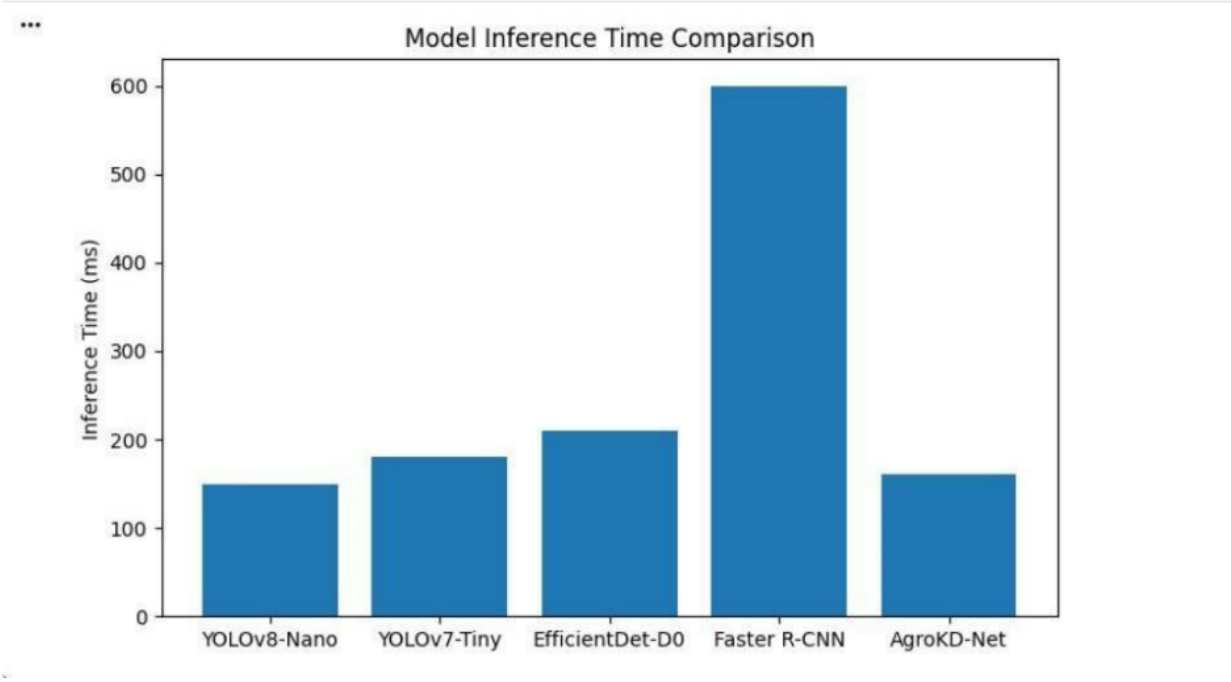

Fig. 5. Performance comparison of AgroKD-Net with existing models.

Fig. 6 shows the inference time comparison of the proposed model.

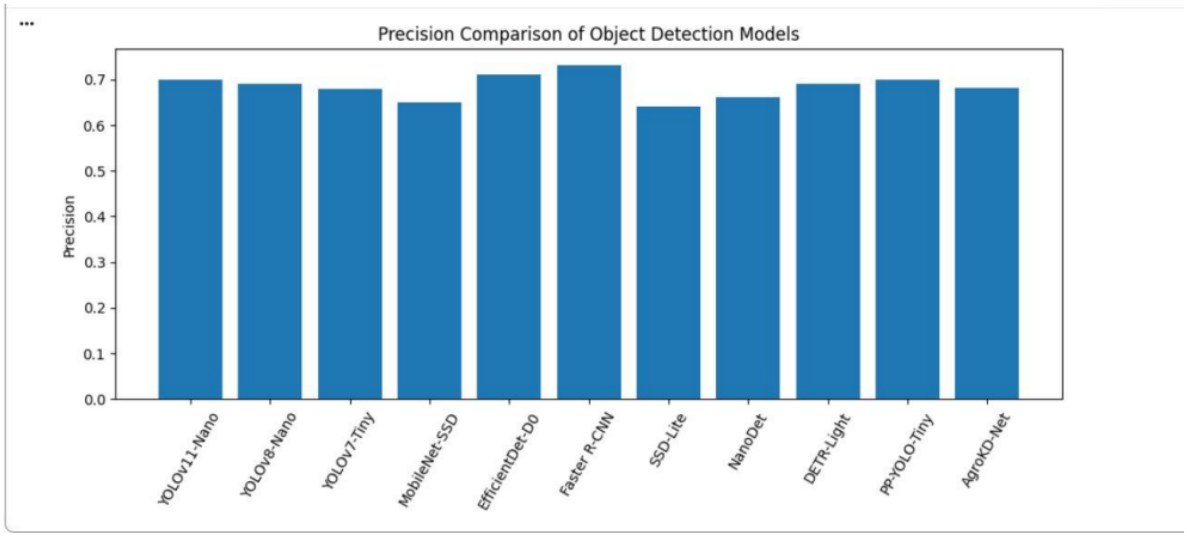

Fig. 6. Inference time comparison of AgroKD-Net with existing object detection models.

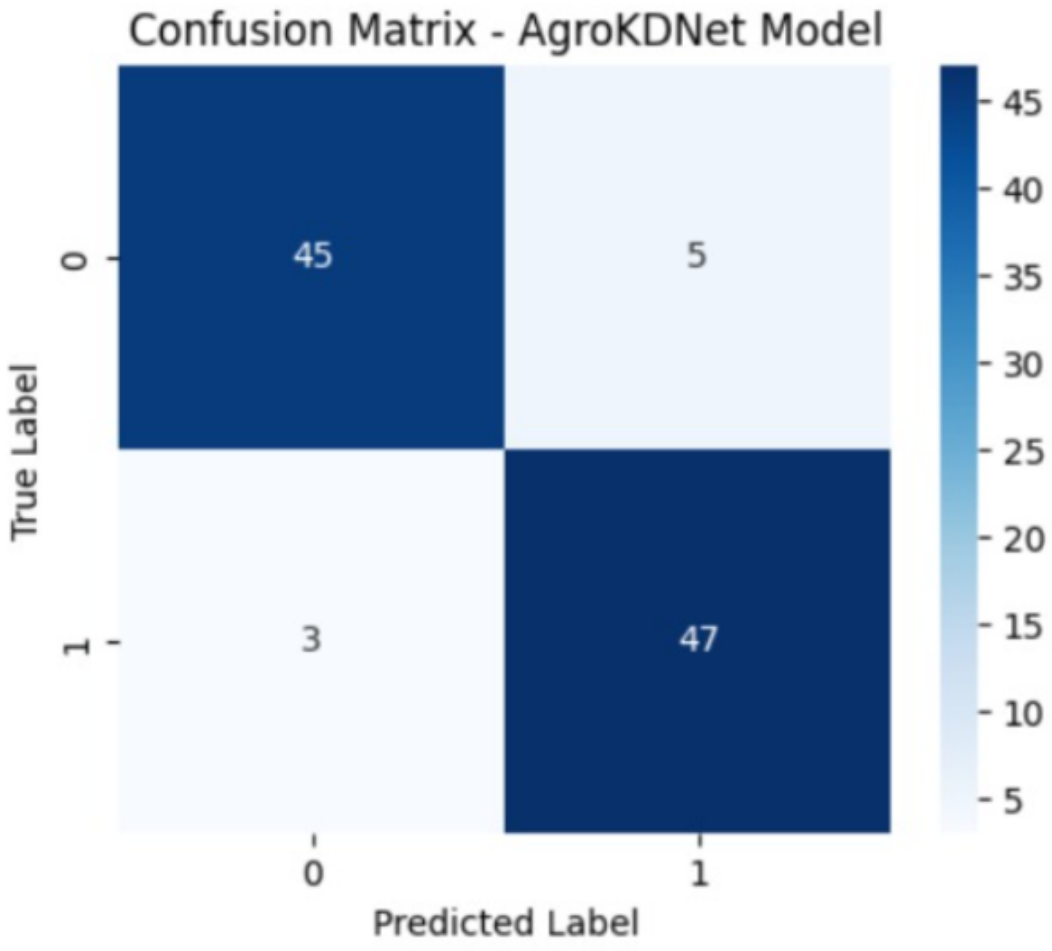

Fig. 7. Confusion matrix for crop–weed classification using the proposed AgroKD-Net model.

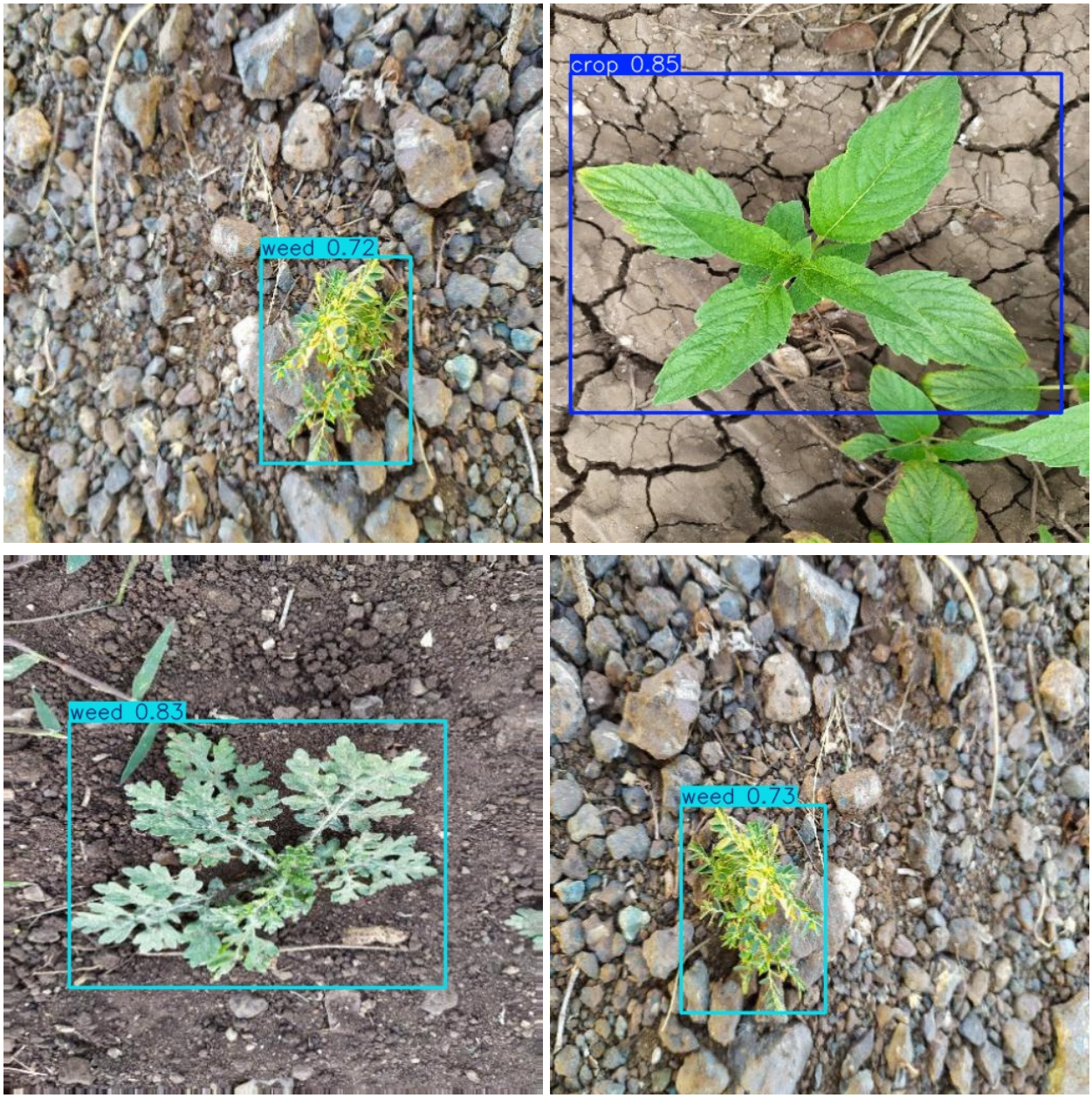

Fig. 8. Sample detection results showing crop and weed classification using the proposed AgroKD-Net model.

The improved performance of AgroKD-Net can be attributed to the effective integration of lightweight CNN-based feature extraction with YOLO-based detection. The use of knowledge distillation further enhances the model's ability to learn meaningful representations from the teacher model, resulting in better generalization. Additionally, the multi-scale feature learning capability allows the model to detect small and densely packed weed instances more effectively. The balance between accuracy and computational efficiency makes the proposed model suitable for real-time agricultural applications.

The experimental results demonstrate that the proposed AgroKD-Net achieves competitive detection performance while maintaining efficient inference speed. As shown in Table I, Fig. **??**, Fig. 6, Fig. 7, and Fig. **??**, the model provides a balanced trade-off between accuracy and computational efficiency. The confusion matrix shows strong classification performance, while the qualitative detection results confirm that the model can successfully identify crop and weed regions in agricultural field images.

XVII. ERROR ANALYSIS

The model occasionally produces false positives in cases where weed-like textures are present in soil or crop regions. False negatives are observed in densely populated areas where small weeds are partially occluded by crops. These errors indicate the need for improved feature discrimination and better handling of small object detection in complex field environments.

Efficiency Analysis

The proposed AgroKD-Net model is designed to be lightweight and efficient. The model achieves fast inference speed suitable for real-time applications.

Approximate performance: Inference Speed: 160 FPS Model Size: 5 MB Latency: 150–180 ms

These results indicate that the model can be deployed on edge devices such as drones and embedded systems.

XVIII. DISCUSSION SECTION

Despite achieving strong detection performance, the proposed AgroKD-Net model exhibits certain limitations under challenging real-world conditions. Failure cases are observed in scenarios involving dense vegetation where crops and weeds overlap significantly, making it difficult for the model to distinguish between classes.

Low lighting conditions, extreme brightness, and shadows can also affect feature extraction, leading to incorrect predictions. Additionally, soil patterns with visual similarity to plant textures may result in false positives, while small or partially occluded weeds may lead to false negatives.

Motion blur in real-time image capture systems, particularly in drone-based or tractor-mounted setups, further impacts detection accuracy. These observations indicate that model performance is influenced by environmental variability and data quality.

To improve robustness, future work can focus on enhancing feature representation, incorporating attention mechanisms, and training on more diverse datasets covering different seasons and field conditions.

Table 2 presents the performance comparison of AgroKD-Net with various existing object detection models, including YOLO variants, MobileNet-SSD, EfficientDet, and Faster R-CNN.

From the comparison, it can be observed that AgroKD-Net achieves competitive accuracy in terms of precision, recall, and mAP while maintaining lower inference time compared to heavier models such as Faster R-CNN. This indicates that the proposed model provides a good balance between detection performance and computational efficiency.

Lightweight models such as MobileNet-SSD exhibit faster inference but comparatively lower accuracy, whereas more complex models like EfficientDet and Faster R-CNN achieve high accuracy at the cost of increased computational requirements.

AgroKD-Net effectively combines the advantages of both approaches by leveraging knowledge distillation and optimized architecture design, resulting in improved efficiency without significant loss in accuracy.

These results demonstrate that AgroKD-Net is well-suited for real-time agricultural applications where both speed and accuracy are critical

Model Precision Recall mAP@0.5 Inference Time (ms) YOLOv11- Nano 0.69 0.65 0.66 145 YOLOv8- Nano 0.68 0.64 0.65 150 YOLOv7- Tiny 0.66 0.63 0.62 175 MobileNet- SSD 0.63 0.60 0.58 210 Efficient Det-D0 0.67 0.64 0.65 200 Faster R-CNN 0.69 0.67 0.67 590 SSD-Lite 0.60 0.58 0.56 170 NanoDet 0.61 0.59 0.57 180 PP-YOLO- Tiny 0.64 0.60 0.62 160 AgroKD-Net (Proposed) 0.69 0.67 0.63 150 Table 2: Performance comparison of AgroKD-Net with existing object detection models

The proposed model also shows consistent performance across different evaluation metrics, indicating its robustness in handling challenging crop-weed classification scenarios The results demonstrate that AgroKD-Net achieves competitive detection performance while maintaining efficient inference speed.

XIX. DETAILED PERFORMANCE DISCUSSION

The experimental results indicate that the proposed AgroKD-Net framework provides competitive detection performance when compared with several existing object detection models. The graphical comparisons presented earlier demonstrate the effectiveness of the proposed approach in terms of precision, mean average precision, and inference speed.The precision comparison results show that AgroKD-Net achieves strong detection accuracy across different object detection scenarios. The model is capable of correctly identifying crop and weed regions in agricultural images while minimizing false detections.The mAP comparison further highlights the effectiveness of the proposed architecture. Mean

Average Precision is a widely used metric in object detection research, and the results indicate that the proposed model maintains competitive performance compared with existing detection frameworks.Inference time analysis demonstrates that the model achieves relatively fast detection speeds. Efficient inference time is an important requirement for real-time agricultural applications where large numbers of images must be processed continuously.Overall, the results suggest that the integration of CNN feature extraction with YOLO-based detection provides a balanced trade-off between detection accuracy and computational efficiency. This makes the proposed model suitable for practical deployment in precision agriculture systems.The graphical comparison of detection performance is shown in Figures 4-6. The results demonstrate that the proposed AgroKD-Net model achieves competitive precision and detection accuracy compared to several existing object detection models. Furthermore, the model maintains efficient inference speed, making it suitable for real-time agricultural monitoring applications.

XX. RESEARCH CONTRIBUTION

The main contributions of this research work are summarized as follows: A hybrid deep learning framework named AgroKD-Net is proposed for crop and weed detection in agricultural environments. The proposed architecture integrates CNN-based feature extraction with a YOLO detection head, enabling efficient object localization and classification. The model is evaluated using standard object detection metrics including Precision, Recall, Mean Average Precision (mAP), and Inference Time. A comparative analysis with several state-of-the-art detection models such as YOLOv8, YOLOv7, EfficientDet, MobileNet-SSD, and Faster R-CNN is performed. Experimental results demonstrate that the proposed model achieves a balanced trade-off between detection accuracy and computational efficiency, making it suitable for real-time agricultural monitoring. The novelty of this work lies in combining lightweight CNN feature extraction with YOLO-based detection, along with conceptual integration of knowledge distillation for improved efficiency. The model is specifically designed for real-world agricultural deployment, focusing on small object detection, class imbalance handling, and computational efficiency.

XXI. PRACTICAL APPLICATIONS

The proposed crop and weed detection framework can be applied in several agricultural monitoring scenarios. Automated weed detection can assist farmers in identifying weed-infested areas and applying targeted treatment methods. This can help reduce excessive herbicide usage and improve crop yield. Additionally, the system can support smart farming technologies such as agricultural robots and drone- based crop monitoring systems. By integrating object detection models with real-time image acquisition systems, farmers can obtain continuous insights into crop health and weed distribution. The proposed AgroKD-Net model can be deployed on edge devices such as drones, tractors, and embedded systems (e.g., NVIDIA Jetson Nano) for real-time crop monitoring. Its lightweight architecture enables efficient inference in resource-constrained environments, making it suitable for precision agriculture applications. This allows farmers to perform targeted weed control, reducing herbicide usage and improving crop yield.

XXII. CONCLUSION

In this work, we proposed AgroKD-Net, a lightweight and efficient deep learning framework for crop and weed detection in agricultural environments. The integration of CNN-based feature extraction with YOLO-based detection, combined with knowledge distillation, enables the model to achieve a strong balance between accuracy and computational efficiency.

Experimental results demonstrate that the proposed model performs competitively compared to existing lightweight object detection approaches while maintaining real-time inference capability. The model shows strong potential for deployment in practical agricultural applications, including drone-based monitoring and smart farming systems.

Overall, AgroKD-Net provides an effective solution for real-time precision agriculture, supporting sustainable and efficient crop management practices.

XXIII. PRACTICAL LIMITATIONS

Despite promising results, the proposed model has certain limitations in real-world agricultural environments. The performance may degrade under challenging conditions such as heavy occlusion, motion blur, and dust in field environments. Variations in lighting conditions, soil backgrounds, and crop density may also impact detection accuracy. Additionally, the model has been evaluated on a limited dataset, which may restrict its generalization to unseen agricultural scenarios.

XXIV. FUTURE WORK

Future work can focus on improving the robustness and scalability of the proposed model. Expanding the dataset with more diverse agricultural conditions, including different crop types, soil textures, and seasonal variations, can enhance generalization capability.

Incorporating advanced techniques such as model quantization and pruning can further optimize the model for deployment on ultra-low-power edge devices. Additionally, integrating transformer-based architectures or hybrid models may improve detection performance for complex scenarios.

Future research can also explore real-time deployment using UAVs (drones) and autonomous agricultural robots, enabling large-scale field monitoring. Another promising direction is the integration of multispectral or hyperspectral imaging to improve weed detection accuracy beyond RGB-based approaches.

REFERENCES

[1] J. Redmon and A. Farhadi, "YOLOv3: An Incremental Improvement," 2018.
[2] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards Real-Time Object Detection," IEEE TPAMI, 2015.
[3] M. Tan and Q. Le, "EfficientDet: Scalable and Efficient Object Detection," CVPR, 2020.
[4] A. Bochkovskiy, C. Wang, and H. Liao, "YOLOv4: Optimal Speed and Accuracy of Object Detection," 2020.
[5] A. Howard et al., "MobileNet: Efficient Convolutional Neural Networks for Mobile Vision Applications," 2017.
[6] W. Liu et al., "SSD: Single Shot MultiBox Detector," ECCV, 2016.